# AI-BASED BUSINESS LOCATION SELECTOR

A Complete Machine Learning Project Guide for Students

**AI-Based Business Location Selector: Project Workflow**




**Project domain:** Machine Learning, Data Analysis, Geospatial Analytics and Decision Support

This guide explains the project idea, problem statement, dataset design, algorithms, methodology, scoring formula, evaluation, dashboard, implementation plan, deliverables, and viva questions.

Prepared as a student-facing project handout

# 1. Project Overview

**Project title:** AI-Based Business Location Selector Using Machine Learning

Develop a system that compares candidate locations for a new business using factors such as rent, population, competition, customer footfall, accessibility, parking, public transport, purchasing power, and nearby facilities. The system produces a transparent ranking and, when suitable labelled data is available, estimates business demand or sales.

**Example:** An entrepreneur wants to open a café in Bhopal. The application compares several candidate areas, considers likely demand and operating costs, and shows which locations best match the entrepreneur's priorities.

# 2. Problem Statement

Choosing a business location is difficult because high population does not always mean high sales, low rent may come with low customer demand, and competition can indicate either market saturation or a strong existing customer base. Manual decisions may rely on incomplete information. This project combines data analysis, machine learning, and a transparent scoring approach to support more informed location decisions.

# 3. Objectives

• Identify the main factors that influence location suitability for a selected business type.

• Collect, clean, document, and analyze location-based data.

• Explore relationships among rent, population, footfall, competition, and business outcomes.

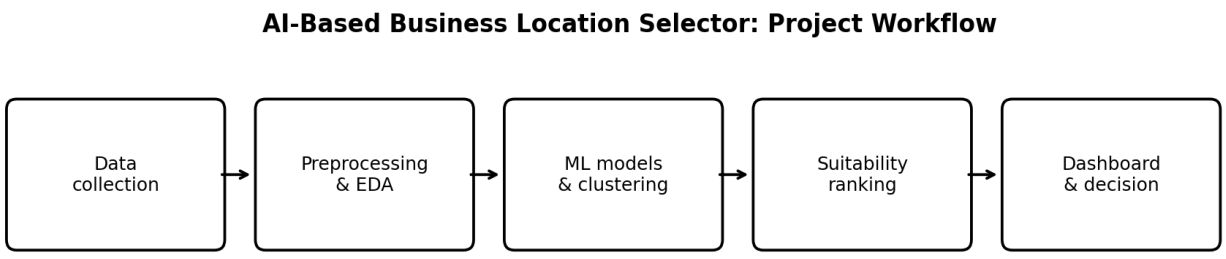
• Train and compare suitable machine learning models for a clearly defined prediction target.

• Use clustering to identify groups of similar candidate locations.

• Create a weighted suitability score and rank candidate locations.

• Build an interactive dashboard that explains the strengths and trade-offs of each recommendation.

# 4. Proposed System Workflow




The system begins with data collection, proceeds through preprocessing and exploratory analysis, applies prediction and clustering models, ranks locations, and presents the results through a dashboard. A map can be added when reliable coordinates are available.

# 5. Features and Dataset Design

A dataset should contain one row per candidate location and columns describing the business type, area, cost, demand indicators, accessibility, competition, and any target variable used for training.

| Feature | Description | Example |
|---|---|---|
| location_name | Candidate area name | MP Nagar |
| business_type | Type of business being evaluated | Cafe |
| monthly_rent | Expected monthly rent in local currency | 35000 |
| population | Estimated population in the chosen catchment area | 25000 |
| competitor_count | Number of similar businesses nearby | 6 |
| daily_footfall | Estimated daily pedestrian flow | 1500 |
| accessibility_score | Ease of reaching the location, e.g. 1–10 | 8 |
| parking_score | Availability/convenience of parking, e.g. 1–10 | 7 |
| public_transport_score | Access to public transport, e.g. 1–10 | 8 |
| income_level | Approximate local purchasing power | Medium |
| nearby_facilities | Nearby offices, schools, colleges, hospitals, etc. | Offices, college |
| target_demand / sales | Observed outcome for supervised model training | Actual value, if available |
| business_potential | Defined category or independently assessed label | Good |

**Data integrity note:** Example values are illustrative. Real data should be sourced and documented. If students generate values or labels, they must identify them as synthetic; a model trained on synthetic labels demonstrates a workflow, not verified real-world predictive accuracy.

# 6. Understanding the Important Features

• **Population:** A larger population can indicate a larger potential customer base, but the relevant catchment area and customer segment matter.

• **Rent:** Lower rent can improve affordability, but a low-cost location may not attract enough customers.

• **Competition:** Many competitors can indicate market saturation, but they may also signal established demand. Interpret this feature alongside other evidence.

• **Footfall:** Pedestrian traffic can matter for cafés and retail stores; appointment-based or delivery businesses may depend on different signals.

• **Accessibility:** Road access, public transport, walkability, and parking can influence how easily customers reach the site.

• **Purchasing power:** Local income patterns can help assess whether a target customer group can afford the product or service.

• **Nearby facilities:** Colleges, offices, hospitals, and residential areas can generate different kinds of demand.

# 7. Machine Learning Algorithms

For an Advanced ML project, students should compare multiple approaches and explain what each one contributes.

## A. Random Forest Regressor

Predict a numerical target such as daily customer count, monthly sales, or a validated demand score. Random Forest combines many decision trees and can model non-linear relationships. It requires a meaningful target variable and appropriate training data.

## B. Decision Tree

Provide interpretable rules and a baseline model. A tree can help explain patterns such as the interaction between rent, footfall, and accessibility. Trees can overfit if they are not controlled.

## C. K-Means Clustering

Group locations with similar characteristics—for example, affordable/high-footfall, premium/high-demand, or low-footfall/high-cost groups. Clustering is unsupervised: it discovers groups but does not prove that a cluster is profitable.

## D. Linear Regression

Use as a simple regression baseline. Compare its performance with Random Forest and explain whether the added complexity improves results on held-out data.

## E. Weighted Suitability Score

Create a transparent decision score from normalized factors. This is a scoring method rather than a trained ML model. It is useful even when there is not enough reliable historical data to train a predictive model.

# 8. Methodology: Step by Step

**Step 1 — Define scope:** Select a business type (such as café, pharmacy, grocery store, or clothing shop), geographic area, candidate locations, and the decision to support.

**Step 2 — Collect data:** Use documented rental estimates, population estimates, points of interest, transport/accessibility indicators, competitor counts, and historical outcomes if available.

**Step 3 — Clean the data:** Check missing values, duplicates, impossible values, inconsistent units, and outdated records. Record the source and date of each dataset.

**Step 4 — Perform EDA:** Use histograms, box plots, scatter plots, correlation analysis, and maps to understand distributions and potential relationships.

**Step 5 — Engineer features:** Calculate competitor density, rent affordability relative to budget, nearby facility counts, and catchment-based indicators. Avoid using future information that would not be available at decision time.

**Step 6 — Split the data:** Create training and test sets. If observations are spatially or temporally related, consider location-based or time-based validation to reduce leakage.

**Step 7 — Train and compare models:** Establish a baseline, train suitable models, tune hyperparameters with cross-validation, and compare test-set results.

**Step 8 — Cluster locations:** Scale appropriate features, run K-Means, evaluate candidate cluster counts with the Elbow Method and silhouette score, and interpret each cluster.

**Step 9 — Rank candidates:** Calculate a transparent score using business-specific weights. Show the score components and trade-offs rather than only the final rank.

**Step 10 — Build and test the dashboard:** Allow users to choose a business type, set priorities, compare candidates, view charts/maps, and export results.

# 9. Suitability Score Formula

A café example can use the following illustrative weighted formula:

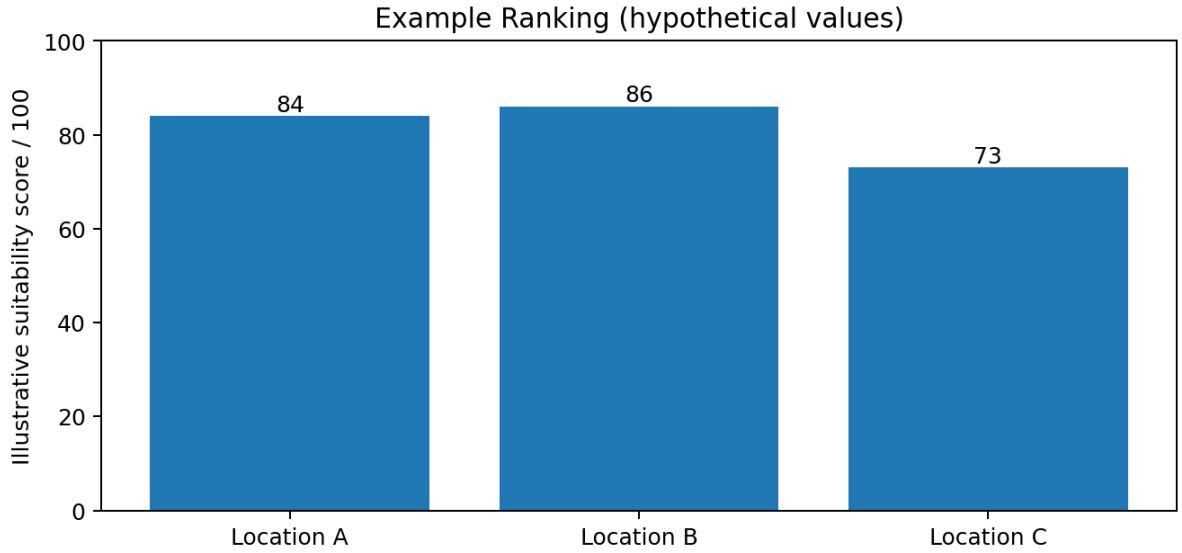
**S = 0.30D + 0.20A + 0.15F + 0.15P + 0.10C + 0.10R**

All components are normalized to a score from 0 to 100. D = customer-demand indicator; A = accessibility; F = footfall; P = parking/public transport; C = competition opportunity (lower saturation may score higher, but context matters); R = rent affordability (rent relative to the business budget). Weights sum to 1.00, so the score ranges from 0 to 100.

These weights are only an example. Students should justify them with stakeholder input, documented evidence, or sensitivity analysis. Different business types need different weights.

## Example hypothetical scores:

| Candidate | Demand | Accessibility | Rent affordability | Final score |
|---|---|---|---|---|
| Location A | 90 | 85 | 65 | 84 |
| Location B | 80 | 90 | 90 | 86 |
| Location C | 95 | 70 | 40 | 73 |




These values are hypothetical examples. Under this example, Location B ranks first because it balances demand, accessibility, and affordability. A score is not a guarantee of profit or proof that a location is commercially successful.

# 10. Dataset Sources and Options

Choose one of these paths depending on the time and data access available.

**Option A — Synthetic dataset:** Generate 500–2,000 candidate-location records in Python. This is easiest for building a complete classroom prototype. Make the synthetic nature of the data explicit and avoid claiming real-world accuracy.

**Option B — Geographic and population data:** WorldPop (https://www.worldpop.org/) provides population estimates; OpenStreetMap (https://www.openstreetmap.org/) includes roads and points of interest. Check data licences, coverage, and update dates before use.

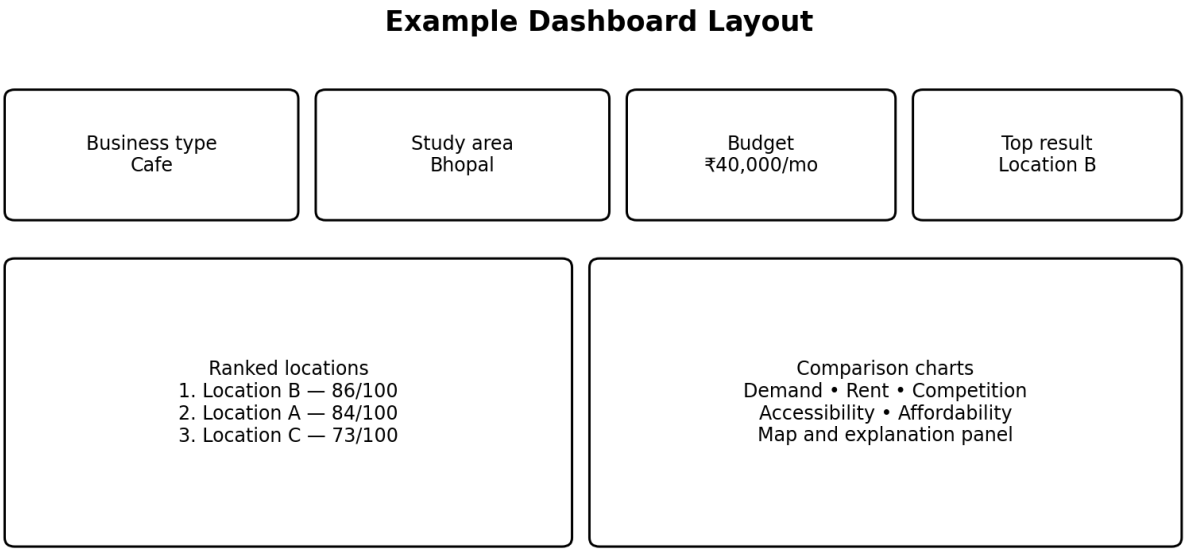
**Option C — Retail transactions:** UCI Online Retail II (https://archive.ics.uci.edu/dataset/502/online+retail+ii) and the UCI Sales Transactions Dataset Weekly (https://archive.ics.uci.edu/dataset/396/sales+transactions+dataset+weekly) can support sales analysis or demand-modelling exercises. These do not by themselves contain candidate-location information, so they must be supplemented with location features.

**Recommended plan:** Start with synthetic data to validate the software workflow. Then enrich the data with documented geographic information and rental estimates. If possible, obtain actual historical sales or customer-count outcomes for meaningful supervised evaluation.

# 11. Software and Libraries

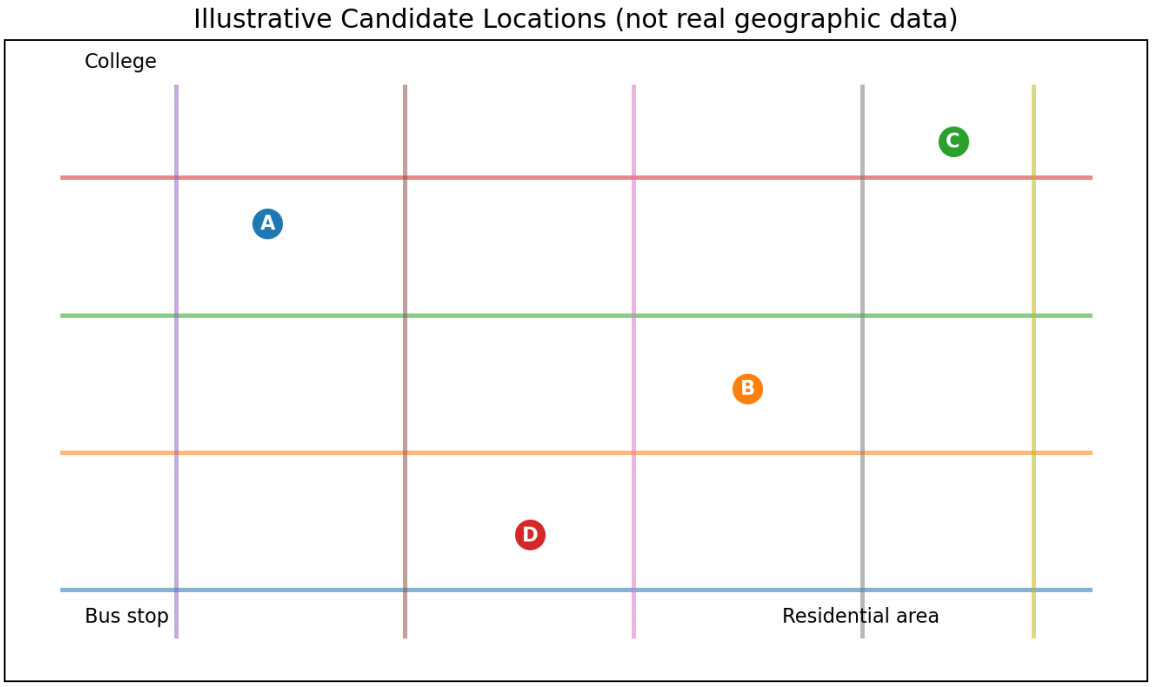
| Tool / library | Purpose |
|---|---|
| Python | Main programming language |
| Jupyter Notebook / Google Colab | Development and experimentation |
| Pandas and NumPy | Data manipulation and numerical calculations |
| Matplotlib and Seaborn | Exploratory charts and visualizations |
| Scikit-learn | Preprocessing, regression/classification, clustering, evaluation |
| Plotly | Interactive charts |
| Streamlit | Interactive application/dashboard |
| Folium | Map visualization when coordinates are available |
| Joblib | Saving and loading trained models |
| SQL (optional) | Storing and querying location records |

# 12. Expected Dashboard




Recommended dashboard sections: business category and budget selector; candidate comparison table; top-five ranking; demand prediction if supported by data; cluster visualization; location map; explanation panel; and CSV/Excel export.

# 13. Example Location Map




The map is a schematic illustration only. In the actual application, plot real coordinates from a documented source. Use catchment areas and travel-time estimates where appropriate, and avoid presenting a schematic map as real geographic evidence.

# 14. Model Evaluation

Evaluate the predictive model separately from the suitability ranking. The correct metric depends on the target variable and how the model will be used.

| Task | Possible metrics | Interpretation |
|---|---|---|
| Numerical demand/sales prediction | MAE, RMSE, $R^2$ | Error magnitude and explained variation on held-out data |
| Suitability category classification | Precision, recall, F1-score, accuracy | Correctness of predicted classes; inspect class imbalance |
| Clustering | Silhouette score and cluster interpretation | Separation of clusters plus whether groups make practical sense |
| Location ranking | Expert review or later observed outcomes | Whether the order is useful and agrees with independent evidence |

A high suitability score is not the same as high predictive accuracy. If there are no observed demand, sales, or independently verified suitability labels, students must not claim that the model predicts real business success. Synthetic targets can only demonstrate the workflow.

# 15. Suggested Four-Week Timeline

| Week | Activities |
|---|---|
| 1 — Plan and prepare | Choose business type and study area; collect or generate data; clean data; perform exploratory analysis. |
| 2 — Build ML models | Train and compare at least two suitable predictive models; implement K-Means; design the scoring formula; evaluate results. |
| 3 — Develop application | Build Streamlit interface; add ranking table, charts, explanations, and optional map. |
| 4 — Test and present | Test unseen data; document limitations; prepare report, slides, screenshots, and demonstration. |

# 16. Final Deliverables Checklist

- Project synopsis and problem statement.
- Literature review and analysis of existing approaches.
- Dataset in CSV/Excel format with source, date, licence, and field descriptions.
- Data-cleaning and exploratory-analysis notebook.
- Model training, comparison, and evaluation notebook.
- Comparison of at least two predictive models where supervised prediction is attempted.
- K-Means clustering analysis with interpretation.
- Suitability scoring and ranking module.
- Working Streamlit application and screenshots.
- Project report with methodology, results, limitations, and references.
- PowerPoint presentation, source code, and README with setup instructions.

# 17. Viva Questions

**1.** What is the objective of an AI-Based Business Location Selector?

**2.** Why is machine learning useful for location analysis?

**3.** What is the difference between regression and classification?

**4.** Why might Random Forest outperform Linear Regression on some datasets?

**5.** What is K-Means clustering, and how does it help here?

**6.** What is the Elbow Method?

**7.** Why may numerical features need scaling before K-Means?

**8.** How can missing values affect results?

**9.** What is the difference between training and test data?

**10.** Which metrics are suitable for regression, and why?

**11.** Why can a high-population area still be unsuitable?

**12.** How can competition indicate both opportunity and risk?

**13.** How would recommendations differ for a café and a pharmacy?

**14.** What is the difference between a suitability score and predicted sales?

**15.** What are the limitations of synthetic data?

**16.** What is overfitting, and how can it be reduced?

**17.** How can geographic information be incorporated?

**18.** How can recommendations be validated in practice?

**19.** How would you explain an individual recommendation to a business owner?

**20.** What features would you add in a future version?

# 18. Advanced Extensions

• Business-specific models or weights for different industries.

• Geospatial competitor density within a radius or travel-time catchment.

• Hyperparameter tuning and cross-validation.

• Explainable AI using feature importance or SHAP, where appropriate.

• Sensitivity analysis for changes in rent, budget, footfall, and competition.

• Scenario comparison for low-budget versus high-footfall strategies.

• Real-world validation using historical outcomes or independent expert assessment.

# 19. Student Assignment Brief

**Objective:** Develop a machine learning application that evaluates and ranks candidate business locations based on demand, rent, competition, population, accessibility, and other relevant factors.

**Instructions:** (1) Select a business category and study area. (2) Collect a dataset or create a documented synthetic dataset. (3) Perform data cleaning, EDA, and feature engineering. (4) Implement and compare at least two suitable ML models. (5) Apply K-Means clustering. (6) Create a suitability scoring and ranking mechanism. (7) Build an interactive Streamlit dashboard. (8) Present the top five candidates with visual explanations. (9) Evaluate the model and discuss limitations. (10) Submit the dataset, source code, notebooks, report, presentation, and setup instructions.

**Expected outcome:** A working prototype that helps users compare candidate locations and make data-informed decisions. It should communicate uncertainty, explain its ranking, and avoid guaranteeing business success.

# 20. Responsible Use and Limitations

• Location data can be incomplete, outdated, or biased toward areas with better data coverage.

• Population and footfall are estimates and should use clearly defined geographic catchments and time periods.

• Competition counts depend on how a competitor is defined and how far away it is.

• Business performance also depends on pricing, product quality, marketing, management, and changing economic conditions.

• Do not use sensitive personal data. Respect data licences and privacy rules.

• Describe the system as decision support, not as a guarantee of profitability or a substitute for due diligence.

# Suggested final project title for submission

**“AI-Based Business Location Selector Using Machine Learning, Clustering, and Multi-Criteria Decision Analysis”**

End of project guide